ADVERSE DRUG REACTIONS (ADRs) AND PHARMACOVIGILANCE

Internship Task Report

1. Aim and Objective

Tounderstand Adverse Drug Reactions (ADRs), classify their major types, analyze potential drug-related reactions using symptoms, medication history and timelines, identify the suspected drug, understand causality assessment, and review ADR reporting and pharmacovigilance practices.

2. Introduction

AnAdverseDrugReaction (ADR) is an undesirable and unintended effect associated with the use of a medicine. ADRs are important in clinical practice because they may result from predictable pharmacological effects, toxicity, immune responses, genetic variation, or effects on fetal development. The historical thalidomide tragedy demonstrated the importance of systematic medicine-safety monitoring.

3. Classification of ADRs

Type	Description	Example
Pharmacological	Predictable and generally dose-dependent response.	Hypotension with calcium-channel blockers.
Pathological	Toxic effect or tissue/organ injury, including effects of toxic metaLbivoelirt edsa.	mage following paracetamol overdose.
Genotoxic	Damage to genetic material or DNA.	Genotoxic effects associated with some anticancer drugs.
Allergic / Hypersensitivity	Immune-mediated reaction; Types I–IV may produce immediatel morm deedlaiayted a enfafepchtysl.	axis or delayed drug hypersensitivity.
Idiosyncratic	Unpredictable reaction in a susceptible individual.	Unusual individual response to a medicine.
Pharmacogenetic	Drug response varies because of genetic differences.	Primaquine-induced hemolysis in G6PD deficiency.
Teratogenic	Drug-related harm to fetal development.	Fetal abnormalities associated with certain medicines.

4. Allergic / Hypersensitivity Reactions

Drughypersensitivityreactionsareimmune-mediated and are commonly described as Types I–IV. The clinical presentation can range from immediate reactions such as anaphylaxis to delayed hypersensitivity reactions. Recognition of the timing, symptoms and clinical pattern is important when evaluating a suspected ADR.

5. Teratogenic Effects and Pregnancy Risk Categories

Thevideo describes the traditional pregnancy-risk categories A, B, C, Dand X. These categories are summarized below as presented in the educational material. Modern regulatory systems may use different pregnancy-labeling approaches, so the table should be understood as a summary of the video rather than current prescribing guidance.

Category	General description in the video	Example
A	Safest/lowest-risk category based on the evidence described.	Levothyroxine
B	Comparatively safer; evidence may include reassuring animal or human data with limitat .	Amoxicillin
C	Animal studies may indicate fetal risk, while adequate human evidence is lacking or limit.	Fluconazole
D	Evidence of fetal risk exists, but clinical benefits may outweigh risks in selected situationPs.	Phenytoin
X	Highest-risk category; fetal risk is considered to outweigh any potential benefit.	Thalidomide

6. Patient Case Analysis for Potential ADRs

For each patient case, the suspected ADR should be assessed systematically rather than assuming that a symptom occurring after medicine exposure was necessarily caused by the medicine.

Step	Information to review	Interpretation
1. Symptoms	Reaction, onset, severity and relevant findings.	Defines the suspected adverse event.
2. Medication history	All current/recent medicines, dose, route, new starts and dose changes.	Identifies possible offending medicines.
3. Timeline	Time from exposure to onset of symptoms.	Establishes temporal relationship.
4. Alternative causes.	Underlying disease, infection, other medicines and other explanations.	Tests whether another cause is more plausible.
5. Dechallenge	Improvement after withdrawal, when clinically appropriate.	Can strengthen the drug-related hypothesis.
6. Suspected drug	Medicine with the strongest overall evidence.	Identifies the candidate medicine.
7. Causality	WHO-UMC, Naranjo or other structured method.	Assesses strength of drug–reaction association.

7. Causality Assessment

The video identifies the Bradford Hill Criteria, WHO-UMC Scale and Naranjo Algorithm as approaches for assessing whether a medicine is responsible for an ADR. Key considerations include temporal association, biological plausibility, known effects of the medicine, alternative causes and response after withdrawal. Rechallenge should not be deliberately performed when it could place a patient at risk.

8. Pharmacovigilance

Pharmacovigilance is the science and activities concerned with detecting, assessing, understanding and preventing adverse effects or other medicine-related problems. It supports continuous medicine-safety surveillance and helps protect patients.

9. ADR Reporting in India

The video describes reporting suspected ADRs through the Pharmacovigilance Programme of India (PvPI), including the ADR PvPI application and official reporting forms. Reports include relevant patient information while protecting confidentiality, suspected medicine details, action taken, and the seriousness of the reaction. Reports can be submitted through appropriate ADR Monitoring Centres (AMCs) and the applicable national reporting system.

10. Practical ADR Reporting Information

When completing a suspected ADR report, use patient initials rather than unnecessary identifying information, provide relevant clinical details, document the suspected medicine and other medicines being taken, describe the reaction and its seriousness, and record actions taken. The report should then be submitted through the appropriate PvPI/CDSCO/AMC reporting route.

11. Key Learning Outcomes

- Understand the definition and major classifications of ADRs.
- Recognize predictable, toxic, immune-mediated, pharmacogenetic and teratogenic reactions.
- Analyze patient symptoms, medication history and timelines before assigning causality.
- Identify a suspected drug using clinical evidence rather than temporal association alone.
- Understand WHO-UMC, Naranjo and Bradford Hill approaches to causality assessment.
- Understand the role of PvPI, ADR Monitoring Centres and ADR reporting in India.

12. Conclusion

Identification of ADRs requires systematic evaluation of symptoms, medication history, timing of exposure and alternative causes. Causality assessment provides a structured method for judging the strength of the drug–reaction relationship. Pharmacovigilance and timely ADR reporting are essential for improving medicine

safety, supporting rational drug use and protecting patients.

13. Source and Scope Note

Thisreport was prepared from the videosummary, examples and timestamps supplied for the internship task. The supplied material explains ADR concepts, causality assessment, pregnancy-risk categories and reporting procedures but does not provide complete individual patient case histories with specific symptoms, medication lists and timelines. Therefore, no patient-specific ADR or suspected drug has been fabricated.